

SHIRA, Russian Federation

WMO: 297560

Lat: 54.500N

Lon: 89.933E

Elev: 1556

StdP: 13.89

Time Zone: 7.00 (E07)

Period: 86-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-22.1	-32.9	0.9	-27.4	-27.5	1.3	-22.0	22.8	11.7	20.5	13.5	0.9	100	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	83.9	63.9	80.0	62.7	76.6	61.1	66.1	77.6	64.5	75.9	62.9	73.6	8.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	87.3	70.8	60.2	82.4	69.8	58.7	78.2	68.6	31.7	77.7	30.4	75.3	29.3	73.4	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.6	15.2	DB	-30.5	89.4	6.1	3.2	-34.9	91.7	-38.4	93.5	-41.9	95.3	-46.3	97.6	(4)
(5)			WB	-30.7	69.2	6.0	3.0	-35.1	71.3	-38.6	73.1	-42.0	74.7	-46.4	76.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.5	1.7	8.9	20.3	36.7	50.9	59.8	64.5	60.9	47.5	35.7	16.2	9.2		
	DBStd	24.01	14.53	12.30	12.75	8.72	10.28	7.32	5.72	6.22	7.64	8.71	13.92	12.57		
	HDD50	6973	1497	1151	922	403	123	12	0	3	135	446	1015	1266		
	HDD65	11261	1962	1571	1387	848	443	184	79	157	526	908	1465	1731		
	CDD50	1315	0	0	0	5	151	306	451	341	59	2	0	0		
	CDD65	130	0	0	0	0	6	28	65	30	1	0	0	0		
	CDH74	1328	0	0	0	0	143	285	539	351	10	0	0	0		
	CDH80	314	0	0	0	0	32	56	134	92	0	0	0	0		
	WSAvg	5.3	3.9	4.4	5.1	7.0	6.9	6.0	5.0	4.7	5.0	5.5	5.3	4.7		
	PrecAvg	29.8	1.5	1.3	2.0	2.4	2.5	3.0	3.3	3.2	2.6	3.0	2.7	2.4		
	PrecMax	38.5	4.8	3.5	5.4	4.8	5.7	5.3	6.7	6.3	4.8	6.3	5.4	5.5		
(15) Precipitation	PrecMin	23.0	0.0	0.0	0.6	0.6	0.4	1.1	2.0	0.7	0.6	1.2	1.4	0.4		
	PrecStd	4.0	1.0	0.8	1.2	1.1	1.3	0.9	1.1	1.1	1.1	1.2	1.0	1.3		
	0.4%	DB	34.9	38.8	53.6	68.2	84.7	85.9	88.1	87.0	75.0	61.7	47.6	40.0		
	MCWB	29.0	31.0	40.9	48.9	60.8	63.7	65.6	65.1	58.9	48.4	39.7	33.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	28.1	33.5	46.5	62.7	77.7	80.9	84.0	82.0	68.6	56.9	41.8	33.6		
	MCWB	23.7	27.6	37.8	46.5	58.0	62.4	65.1	63.3	55.6	46.3	34.9	28.3			
	5%	DB	24.7	30.1	41.5	57.4	73.4	77.1	80.0	77.5	64.1	53.1	37.8	29.6		
	MCWB	21.1	25.3	33.9	44.0	55.4	60.2	64.0	62.4	52.9	44.5	31.7	24.9			
	10%	DB	20.9	26.6	37.4	51.9	69.1	73.5	76.4	73.0	60.2	49.1	34.0	26.5		
	MCWB	17.7	22.6	30.9	41.8	53.9	59.3	62.2	60.4	50.6	42.0	29.0	22.6			
	0.4%	WB	29.3	31.4	42.7	50.6	61.2	65.7	70.2	67.5	60.5	50.0	40.0	34.0		
	MCDB	35.6	37.3	52.3	62.6	79.9	79.0	81.5	81.5	71.5	60.1	47.1	40.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	23.9	27.8	37.8	47.3	59.1	63.9	67.0	65.3	56.6	47.3	35.7	28.9		
	MCDB	27.7	33.3	45.9	61.4	75.4	76.6	78.9	77.8	66.2	56.0	42.0	33.7			
	5%	WB	21.0	25.3	34.3	44.6	56.5	62.1	65.3	63.4	54.0	45.1	32.0	25.1		
	MCDB	24.5	30.0	41.3	56.0	71.0	73.5	75.8	75.4	62.1	52.7	37.1	29.3			
	10%	WB	17.7	22.6	31.0	41.9	54.3	60.3	63.8	61.3	51.7	42.2	29.2	22.5		
	MCDB	20.8	26.8	36.7	51.7	67.7	71.2	73.7	71.8	58.5	48.9	34.0	26.5			
	MDBR	14.7	17.4	18.8	20.0	20.4	18.8	17.7	18.3	16.8	15.9	13.8	14.8			
	5% DB	MCDBR	18.6	19.5	22.3	29.2	28.6	24.8	24.5	26.9	24.5	22.1	16.8	17.5		
(36) Mean Daily Temperature Range	MCWBR	15.0	15.1	15.1	17.0	14.9	10.8	9.9	11.5	13.5	14.2	11.8	14.1			
	5% WB	MCDBR	19.4	19.3	21.4	25.7	25.3	22.2	19.7	22.5	21.8	20.9	17.0	17.0		
	MCWBR	15.7	15.3	14.4	15.9	13.8	10.2	9.3	10.5	12.9	14.1	12.4	14.3			
	taub	0.264	0.297	0.347	0.393	0.400	0.397	0.416	0.398	0.359	0.335	0.299	0.265			
(40) Clear-Sky Solar Irradiance	taud	2.090	2.057	2.010	2.141	2.205	2.311	2.273	2.292	2.345	2.279	2.116	2.029			
	Ebn at Noon	210	243	256	262	268	270	261	259	255	230	191	176			
	Edh at Noon	21	31	42	42	42	38	39	36	30	25	21	18			
	RadAvg	292	608	1057	1424	1626	1855	1643	1376	946	512	264	186			
(44) All-Sky Solar Radiation	RadStd	31	53	91	76	195	171	108	126	106	52	36	24			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page